

MSc

Course title: Bayes Methods
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1. An 18×24 square-foot area of grass in a field was split into 3×3 square-foot cells (so $n = 48$). An indicator for army-worm damage was recorded for the centre square in each cell. When the grass was planted the plough and harrows ran down the columns.

Northing	Easting							
	1	2	3	4	5	6	7	8
1	1	1	1	0	0	0	0	1
2	1	1	1	0	0	1	0	1
3	0	1	1	1	1	0	1	1
4	1	1	1	0	0	1	1	1
5	1	1	0	0	0	0	0	1
6	0	0	1	1	0	1	1	1

For $i = 1, \dots, n$, let $Y_i \in \{0, 1\}$ indicate army-worm damage observed in cell i . The data are displayed above. Suppose $Y_i \sim \text{Bernoulli}(\mu_i)$ independently in each cell, with

$$\mu_i = \frac{\exp(z_i)}{1 + \exp(z_i)}.$$

where $z_i \in \mathbb{R}$ is an unobserved real latent variable associated with cell i . Let $z = (z_1, \dots, z_n)$. The z -values are modelled by an auto-regression with an $n \times n$ weight matrix W

$$z = (I - W)^{-1}\epsilon, \quad \epsilon \sim N(0, \sigma^2 I_n)$$

It is of interest to estimate a map of the log-odds for damage, and to test for an effect due to ploughing. Assume a prior for σ is available.

(a) [9 marks] Consider eliciting a prior for z by specifying W .

- (i) Suppose that for $i = 1, \dots, n$, $W_{ii} = 0$. Let $\pi(z_i | z_{-i}, \sigma)$ denote the conditional density of z_i given the other z -values and σ . Show that

$$p(z_i | z_{-i}, \sigma) = N(z_i; \sum_{j \neq i} W_{i,j} z_j, \sigma^2).$$

Solution: (2, S) Since $z - Wz = \epsilon$ and $W_{ii} = 0$, we have for each $i = 1, \dots, n$ that $z_i = \sum_{j \neq i} W_{i,j} z_j + \epsilon_i$. The conditional density of z_i is then normal mean $\sum_{j \neq i} W_{i,j} z_j$ and variance σ^2 .

- (ii) Scientist A believes a small number of army worm midges blew in after planting, so that the first arrivals were scattered at random over the area and these then expanded through the ground to form local clusters. Using the auto-regression, elicit a prior $\pi_A(z | \sigma, u)$ for z with a single scalar parameter u .

Solution: (3, S) Let $i \sim j$ if i is a neighbor of j on the lattice. The cells are all equally spaced and there is nothing to distinguish easting and northing. Also, we would expect some sort of local Markov structure, since the army-worm spread through the ground. We set $p(z_i | z_{-i}, \sigma) = p(z_i | z_{j \sim i}, \sigma, u)$ with $u > 0$ a coupling parameter appearing in $W^{(A)}$,

$$W_{i,j}^{(A)} = \begin{cases} u & \text{if } i \sim j \text{ and} \\ 0 & \text{otherwise.} \end{cases}$$

Note that $W^{(A)}$ is symmetric since $i \sim j$ is reflexive. This gives

$$\pi_A(z|\sigma, u) = N(z; 0, (I - W^{(A)})^{-1}(I - W^{(A)})^{-1}\sigma^2).$$

Remark: (not required) This is likely to leave a boundary effect in which the prior variance of z_i for i a perimeter cell is larger than that for interior cells. Also there is likely to be some overall anisotropy due to the lattice orientation.

- (iii) Scientist B believes the army worms blew in before planting and formed small clusters a foot or so across which were subsequently smeared out along the columns by ploughing and seeding. Using the auto-regression, elicit a prior $\pi_B(z|\sigma, v)$ for z with a single scalar parameter v .

Solution: (2, S) By the same reasoning we set $p(z_i|z_{-i}, \sigma) = p(z_i|z_{j \sim i}, \sigma, v)$ with $v > 0$ a parameter in $W^{(B)}$,

$$W_{i,j}^{(B)} = \begin{cases} v & \text{if } i \sim j \text{ and } j \text{ is directly north or south of } i \\ 0 & \text{otherwise.} \end{cases}$$

This will make the columns independent in the prior. Note that $W^{(B)}$ is again symmetric. This gives

$$\pi_B(z|\sigma, v) = N(z; 0, (I - W^{(B)})^{-1}(I - W^{(B)})^{-1}\sigma^2).$$

- (iv) Suppose priors for σ , u and v have been elicited. Say how you would check that these priors and the priors for z reflect available knowledge. Indicate any further information you would need from Scientists A and B.

Solution: (2, S) We could simulate synthetic data by simulating $z' \sim \pi_A(z|\sigma, u)$ (or similarly B) and then for $i = 1, \dots, n$ simulate $Y'_i \sim \text{Bernoulli}(\mu_i(z'_i))$. We could check with the scientists that Y' shows plausible variation by estimating for example the correlation of Y_i and Y_j for i and j neighboring cells and i and j distant pairs. We would reference this to their previous experience with army-worm, not to the data in hand. We could also consider model elaboration, incorporating different north and south weights, or additional diagonal weights, and seeing if it influenced our analysis (ie checking conclusions dont change).

- (b) [11 marks] Assume independent priors $f(\sigma)$, $g(u)$ and $h(v)$ are given.
- (i) Let $m = A, B$ indicate the model and let $u = \theta$ if $m = A$ and $v = \theta$ if $m = B$. Elicit a prior for the model index m and write down the posterior distribution $\pi(z, m, \sigma, \theta|y)$ for the model and parameters in as much detail as you can.

Solution: (3, S) Take $\pi_M(m) = 1/2$. This is natural given the small number of models and the balance of prior evidence. Here

$$\pi(z, m, \sigma, \theta|y) \propto \begin{cases} L(z; y)\pi_A(z|\sigma, \theta)f(\sigma)g(\theta)\pi_M(m = A) & \text{if } m = A \text{ and} \\ L(z; y)\pi_B(z|\sigma, \theta)f(\sigma)h(\theta)\pi_M(m = B) & \text{if } m = B \end{cases}$$

with $\pi_m(z|\sigma, \theta)$ given above for $m = A, B$ and

$$L(z; y) = \prod_{i=1}^n \mu_i(z_i)^{y_i} (1 - \mu_i(z_i))^{1-y_i}.$$

The parameter space is $(z, m, \sigma, \theta) \in \Omega$. The parameter space for the two models is the same, $(z, \sigma, u) \in \Omega_0$ with $\Omega_0 = \mathbb{R}^n \times \mathbb{R}^+ \times \mathbb{R}^+$ and $(z, \sigma, v) \in \Omega_0$, so

$$\Omega = \Omega_0 \times \{A, B\}.$$

The marginal likelihood (ie, the normalising constant) is

$$p(y) = \frac{1}{2} \int_{\mathbb{R}^+ \times \mathbb{R}^+} \int_{\mathbb{R}^n} L(z; y) \pi_A(z|\sigma, u) f(\sigma) g(u) dz du d\sigma + \frac{1}{2} \int_{\mathbb{R}^+ \times \mathbb{R}^+} \int_{\mathbb{R}^n} L(z; y) \pi_B(z|\sigma, v) f(\sigma) h(v) dz dv d\sigma.$$

- (ii) Give the z and m updates of an MCMC algorithm targeting $\pi(z, m|y)$. You should give the proposals and acceptance probabilities in as much detail as you can.

Solution: (4, S) We should target $\pi(z, m, \sigma, \theta|y)$ and ignore σ, θ as the marginal $\pi(z, m|y)$ is presumably intractable. (it might not be, but if so they should say). Suppose the current state is z, m, σ, θ .

1. Propose new z . For $i = 1, \dots, n$ set $z'_i = z_i$ and replace entry i with $z'_i \sim N(z_i, \sigma^2)$. Leave all other parameters unchanged. The acceptance probability is

$$\alpha(z'|z) = \min \left\{ 1, \frac{\mu_i(z'_i)^{y_i} (1 - \mu_i(z'_i))^{1-y_i} p_m(z'_i|z_{-i}, \sigma)}{\mu_i(z_i)^{y_i} (1 - \mu_i(z_i))^{1-y_i} p_m(z_i|z_{-i}, \sigma)} \right\}$$

where $p_m(z_i|z_{-i}, \sigma) = N(z_i; \theta \sum_{j \sim_i^m} z_j, \sigma^2)$ where $j \sim_i^m$ uses the neighborhood structure of $W^{(m)}$.

2. Propose a change of model. If $m = A$ set $m' = B$ and *vis versa*. Leave all other parameters unchanged. The acceptance probability is

$$\alpha(m'|m) = \min \left\{ 1, \frac{\pi_{m'}(z|\sigma, \theta)}{\pi_m(z|\sigma, \theta)} \right\}.$$

- (iii) Outline briefly how to tune your MCMC for efficiency and check for convergence.

Solution: (2, S) Make a few short runs and vary details of proposals trying to increase ESS. For convergence inspect output trace (log-likelihood is a good choice) and check for stationarity (visually) and mixing (using ESS and checking autocorrelation functions fall to zero at a lag much less than run length, so ESS estimate is reliable). Make repeat runs from widely varying initial conditions and check posterior distributions of scalar functions of output z agree to within monte carlo error.

- (iv) The MCMC output is $m^{(t)}, z^{(t)}$, $t = 1, \dots, T$. Explain how to do model selection

and how to estimate $E(Z|Y = y)$.

Solution: (2, S) Estimate $\pi(m = A|y)$ using $\hat{\pi}_A = T^{-1} \sum_{t=1}^T \mathbb{I}_{m^{(t)}=A}$. Since the prior puts equal weight on each model, the Bayes factor for A v. B is estimated by $\hat{B}_{A,B} = \hat{\pi}_A / (1 - \hat{\pi}_A)$. The model averaged posterior mean for z is simply $\bar{z} = T^{-1} \sum_{t=1}^T z^{(t)}$ (in terms of vectors).

2. Data on insects caught over several nights in a mesh net are recorded. Let $N_{i,j}$ give the number of insects of species $i = 1, \dots, n$ caught on night $j = 1, \dots, J$. Let $\lambda_{i,j}$ give the arrival rate per night for species i and let β_i and γ_j be random effects for species and night. In a log-linear model $\log(\lambda_{i,j}) = \alpha + \beta_i + \gamma_j$ and $N_{i,j} \sim \text{Poisson}(\lambda_{i,j})$ independently. Let $N = (N_{i,j})_{i=1, \dots, n}^{j=1, \dots, J}$.

It is thought likely that many of the β -values are equal for groups of insects with similar abundance and behavior. Let $\beta^* = (\beta_1^*, \dots, \beta_K^*)$ denote the K distinct β -values (with K unknown). Let $\gamma = (\gamma_1, \dots, \gamma_J)$. The prior distribution for the J random effects is $\gamma \sim N(0, \sigma_g^2 I_J)$. Assume priors $\pi(\sigma, \sigma_g)$ are given.

- (a) [10 marks] In this part we consider a Dirichlet process model for the grouping with $S = (S_1, \dots, S_K)$ a K -set partition of $\{1, \dots, n\}$ having a Chinese Restaurant Process prior distribution $P_\alpha(S)$ with given parameter α determining the distribution of K . If species i and j are in the same set, so that $i, j \in S_k$ for some k in $1, \dots, K$, then $\beta_i = \beta_j = \beta_k^*$.
- (i) Write down, in as much detail as you can, the posterior $\pi(S, \beta^*, \gamma, \sigma, \sigma_g | N)$ with base distribution $\beta^* \sim N(0, \sigma^2 I_K)$ and concentration parameter α .

Solution: (S, 5) Let $\theta = (\gamma, \sigma, \sigma_g)$ and suppose $S = \{S_1, \dots, S_K\}$. The posterior is

$$\pi(S, \beta^*, \theta | N) \propto \pi(\theta) P_\alpha(S) \pi(\beta^* | S, \theta) L(S, \beta^*, \theta; N)$$

where $P_\alpha(S) \propto \alpha^K \prod_k (|S_k| - 1)!$ when $|S_k|$ is the number of elements in S_k ,

$$\pi(\beta^* | S, \theta) = N(\beta^*; 0, \sigma^2 I_K),$$

$\pi(\theta) = \pi(\sigma, \sigma_g) N(\gamma; 0, \sigma_g^2 I_J)$ and

$$L(S, \beta^*, \theta; N) = \prod_{k=1}^K \prod_{i \in S_k} p(N_i | \lambda_k^*)$$

where $\lambda_k^* = (\lambda_{k,1}^*, \dots, \lambda_{k,J}^*)$ with $\lambda_{k,j}^* = \exp(\alpha + \beta_k^* + \gamma_j)$ for $j = 1, \dots, J$ and

$$p(N_i | \lambda_k^*) \propto \prod_{j=1}^J \exp(\lambda_{k,j}^*) (\lambda_{k,j}^*)^{N_{i,j}}.$$

The parameter spaces are $\theta \in \Omega_\theta$ with $\Omega_\theta = \mathbb{R}^+ \times \mathbb{R}^+ \times \mathbb{R}^J$, and $\beta^* \in \Omega_K$ where $\Omega_K = \mathbb{R}^K$. Let $\mathcal{P}_K$ be the set of all distinct K -set partitions of $\{1, \dots, n\}$. The parameter space, $(S, \beta^*, \theta) \in \Omega$, is

$$\Omega = \Omega_\theta \times \bigcup_{K=1}^n \bigcup_{S \in \mathcal{P}_K} \{S\} \times \Omega_K,$$

and the normalising constant for the posterior is

$$p(N) = \sum_{k=1}^n \sum_{S \in \mathcal{P}_k} \int_{\Omega_K} \int_{\Omega_\theta} \pi(\theta) P_\alpha(S) \pi(\beta^* | S, \theta) L(S, \beta^*, \theta; N) d\theta d\beta^*.$$

- (ii) Give the S update for a Gibbs sampler targeting $\pi(S, \beta^*, \gamma, \sigma, \sigma_g | N)$.

Solution: (N, 5) Let $S_{-i} = (S_{-i,1}, \dots, S_{-i,K_{-i}})$ and $\beta_{-i}^* = (\beta_{-i,1}^*, \dots, \beta_{-i,K_{-i}}^*)$ be the partition and the vector of parameters which remain after i is removed. Note that K_{-i} may be less than K if i is in a cluster by itself. In that case we delete the empty cluster and remove the corresponding element of β^* .

We need $P(i \in S_{-i,k} | S_{-i}, \beta_{-i}^*, \theta, N)$. Since the insects $i = 1, \dots, n$ are exchangeable we may think of i as the last arrival, which chooses an existing cluster k w.p. $|S_{-i,k}|/(\alpha + n - 1)$ and a new cluster w.p. $\alpha/(\alpha + n - 1)$. The joint distribution for N_i and k is then

$$P(i \in S_{-i,k}, N_i | S_{-i}, \beta^*, \theta, N_{-i}) \propto \begin{cases} |S_{-i,k}| p(N_i | \lambda_k^*) & \text{if } k \in \{1, \dots, K\} \\ \alpha p(N_i | \lambda_{K_{-i}+1}^*) & \text{if } k = K_{-i} + 1. \end{cases}$$

In the second case, where i is placed in a new cluster $k = K_{-i} + 1$, we simulate $\beta_k^* \sim N(0, \sigma^2)$ and set $\lambda_k^* = (\exp(\alpha + \beta_k^* + \gamma_j))_{j=1, \dots, J}$. Now the conditional distribution is proportional so

$$P(i \in S_{-i,k} | S_{-i}, \beta^*, \theta, N) \propto P(i \in S_{-i,k}, N_i | S_{-i}, \beta^*, \theta, N_{-i}).$$

If $P_i = (P(i \in S_{-i,k} | S_{-i}, \beta^*, \theta, N))_{k=1, \dots, K}$ is the vector of set choice probabilities for i then the S -update visits each $i = 1, \dots, n$ in turn, sets $S \leftarrow S_{-i}$, draws $k \sim V_i$ and sets $S_k \leftarrow S_k \cup \{i\}$, simulating $\beta_k^* \sim N(0, \sigma^2)$ if $k = K_{-i} + 1$.

- (b) [10 marks] In a new model the unknown number of clusters has a Poisson distribution $K \sim \text{Poisson}(M)$ with M fixed. Let $w = (w_1, \dots, w_K)$ be a vector of positive weights, normalised so that $\sum_k w_k = 1$ and having prior $w \sim \text{Dirichlet}(\alpha/M, \dots, \alpha/M)$. Consider the same data, but suppose the observation model is now a mixture,

$$N_{i,j} \sim \sum_{k=1}^K w_k \text{Poisson}(\lambda_{k,j}^*), \text{ independently for } i = 1, \dots, n \text{ and } j = 1, \dots, J,$$

with $\lambda_{k,j}^* = \exp(\alpha + \beta_k^* + \gamma_j)$. The priors for $\beta^* | K$, γ , σ and σ_g are unchanged.

- (i) Write down, in as much detail as you can, the posterior $\pi(w, \beta^*, \gamma, \sigma, \sigma_g | N)$ in terms of the likelihood and parameter priors (note that $K = \dim(\beta^*)$ is known given β^* so we make no separate reference to this parameter).

Solution: (S, 5) Let $\theta = (\gamma, \sigma, \sigma_g)$. The posterior is

$$\pi(w, \beta^*, \theta | N) \propto \pi(\theta) \text{Poisson}(K; M) \pi(\beta^* | K, \theta) \pi(w | K) L(\beta^*, \theta; N)$$

where

$$\pi(w | K) = \frac{\Gamma(K\alpha/M)}{\Gamma(\alpha/M)^K} \prod_{k=1}^K w_k^{\alpha/M-1},$$

$$\pi(\beta^* | K, \theta) = N(\beta^*; 0, \sigma^2 I_K),$$

$$\pi(\theta) = \pi(\sigma, \sigma_g) N(\gamma; 0, \sigma_g^2 I_J) \text{ and}$$

$$L(\beta^*, \theta; N) = \prod_{i=1}^n \prod_{j=1}^J \sum_{k=1}^K w_k p(N_{i,j} | \lambda_{k,j}^*)$$

where $\lambda_{k,j}^* = \exp(\alpha + \beta_k^* + \gamma_j)$ for $j = 1, \dots, J$ and

$$p(N_{i,j} | \lambda_{k,j}^*) \propto \exp(\lambda_{k,j}^*) (\lambda_{k,j}^*)^{N_{i,j}}.$$

The parameter spaces are $\theta \in \Omega_\theta$ with $\Omega_\theta = \mathbb{R}^+ \times \mathbb{R}^+ \times \mathbb{R}^J$, $w \in \Omega_{w,K}$ with $\Omega_{w,K} = \{w \in [0, 1]^K; \sum_k w_k = 1\}$ and $\beta^* \in \Omega_{\beta,K}$ where $\Omega_{\beta,K} = \mathbb{R}^K$. The parameter space, $(w, \beta^*, \theta) \in \Omega$, is

$$\Omega = \Omega_\theta \times \bigcup_{K=1}^{\infty} \Omega_{\beta,K} \times \Omega_{w,K},$$

and the normalising constant for the posterior is

$$p(N) = \sum_{K=1}^{\infty} \int_{\Omega_{\beta,K} \times \Omega_{w,K} \times \Omega_\theta} \pi(\theta) \text{Poisson}(K; M) \pi(\beta^* | K, \theta) \pi(w | K) L(\beta^*, \theta; N) d\theta dw d\beta^*.$$

- (ii) In a reversible jump MCMC algorithm targeting $\pi(w, \beta^*, \gamma, \sigma, \sigma_g | N)$, the proposals to add or drop an element of $\beta^* = (\beta_1^*, \dots, \beta_K^*)$ are chosen with probabilities $\rho_{K,K+1}$ and $\rho_{K,K-1}$ respectively. Give proposals for adding and removing elements of β^* and the acceptance probability for the move to add a dimension. Distributions must be well defined, but there is no need to simplify in acceptance probabilities.

Solution: (N, 5) Let w, β^* denote the old state and w', β' denote the proposal. In order to add a dimensions a new state w', β' is generated by setting $w' = w$, $\beta' = \beta^*$ and then simulating an additional parameter $\beta'_{K+1} \sim N(0, \sigma^2)$. An associated weight $w'_{K+1} \sim U(0, w_i)$ is simulated by choosing $i \sim U\{1, \dots, K\}$ and setting $w'_i = w_i - w'_{K+1}$.

In order to delete a dimension a new state w', β' is generated by choosing an element $i \sim U\{1, \dots, K\}$ at random from β^* and setting $\beta' = \beta^*_{-i}$ (ie removing the i th element). The new weight vector is determined by setting $w' = w_{-i}$, choosing an element $j \sim U\{1, \dots, K\} \setminus \{i\}$, and updating $w'_j = w_j + w_i$. When we add a dimension $K \rightarrow K+1$ the probability distribution to generate the new state from the old is

$$q(w', \beta' | w, \beta^*) dw' d\beta' = \rho_{K,K+1} N(d\beta'_{K+1}; 0, \sigma^2) \times \frac{1}{K} \times \frac{dw_i}{w_i}$$

since we choose the move, generate β'_{K+1} , choose i and slice w_{K+1} off w_i . The probability to generate the old state from the new is

$$q(w, \beta^* | w', \beta') = \rho_{K+1,K} \times \frac{1}{K(K+1)}$$

since we choose the move, choose the element we added as the element to delete from the extended list, and then choose the element we sliced as the element to receive the deleted weight.

The acceptance probability for the proposal to add a dimension is then

$$\alpha(w', \beta' | w, \beta^*) = \left\{ 1, \frac{\pi(w', \beta', \theta | N) q(w, \beta^* | w', \beta')}{\pi(w, \beta^*, \theta | N) q(w', \beta' | w, \beta^*)} \right\}$$

in terms of the distributions q and $\pi(w', \beta', \theta | N)$ above. This is tractable as the normalisation cancels. This simplifies very nicely but I do not expect further processing.